SOC Operations, Detection Engineering & Incident Response Capstone Report

GitHub Repository: `ace138/soc-capstone-project` | **Date:** October 2026

1. Lab Architecture & Log Ingestion Verification (Step 1.1)

Role	Hostname	IP Address	OS / Platform	Telemetry & Verified Latency
SIEM Server	`splunk-siem`	`192.168.120.149`	Ubuntu / Splunk Enterprise (`:8000`)	Central Indexing, Correlation & Alerting (TCP `9997`)
Windows Endpoint	`DESKTOP-DTUQ6D4`	`192.168.120.150`	Windows 10 Enterprise	`WinEventLog:Security` (`4688, 4624, 4625, 4698, 1102`), `System` (`7045, 104`), `PowerShell` (`4104`) — **1.0s Delay**
Linux Endpoint	`web`	`192.168.120.151`	Ubuntu Server (`auditd`)	`sourcetype="linux:audit"` (`cron_persistence`, `ssh_keys`) — **0.638s Delay**

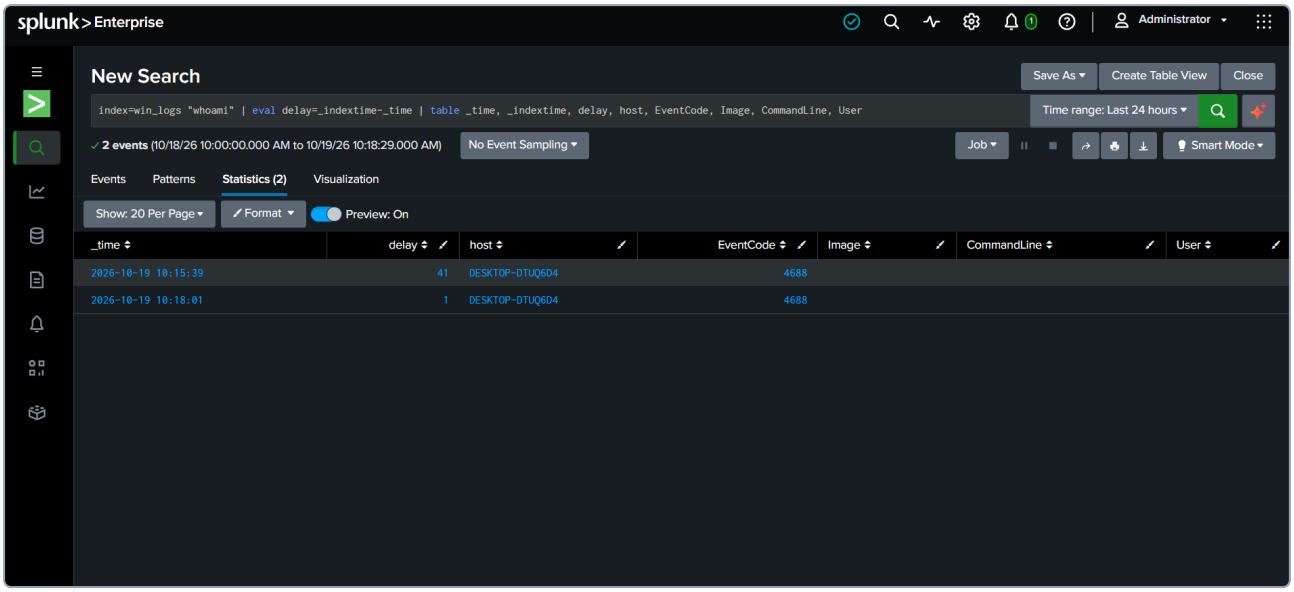


Figure 1.1a: Windows Endpoint (DESKTOP-DTUQ6D4) EventCode=4688 Ingestion & Sub-Second Latency Verification

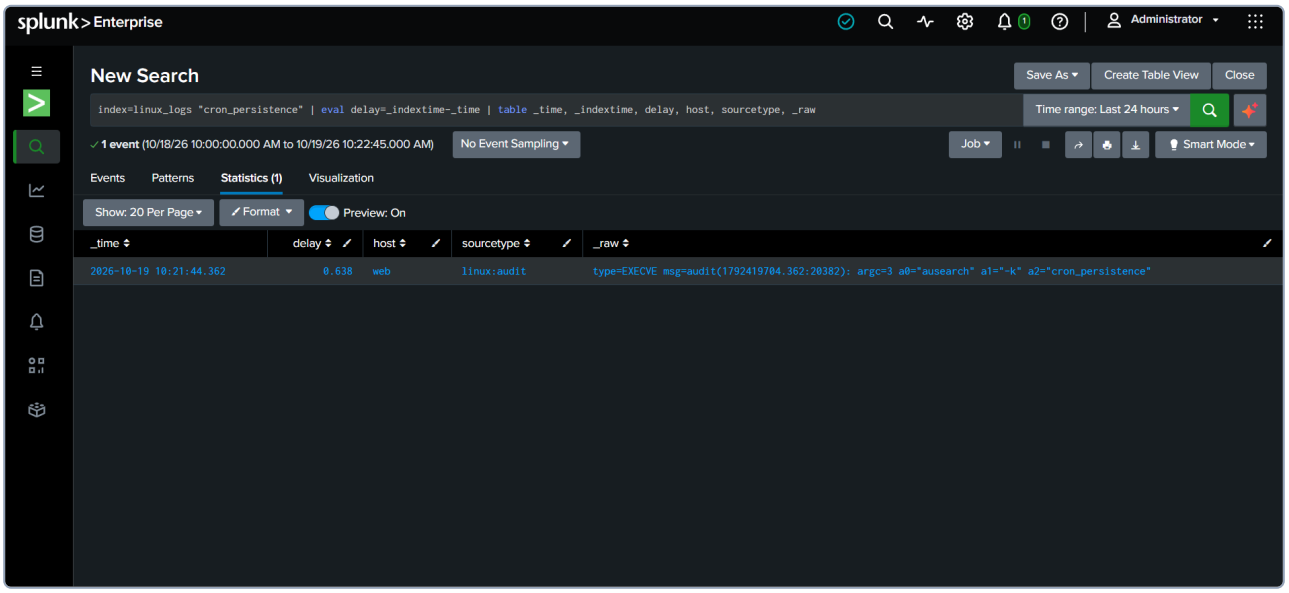


Figure 1.1b: Linux Endpoint (web) Auditd Syscall Ingestion & 0.638s Latency Verification

2. Detection Engineering: 10 Custom Rules & Alert Validation (Steps 1.2–1.3)

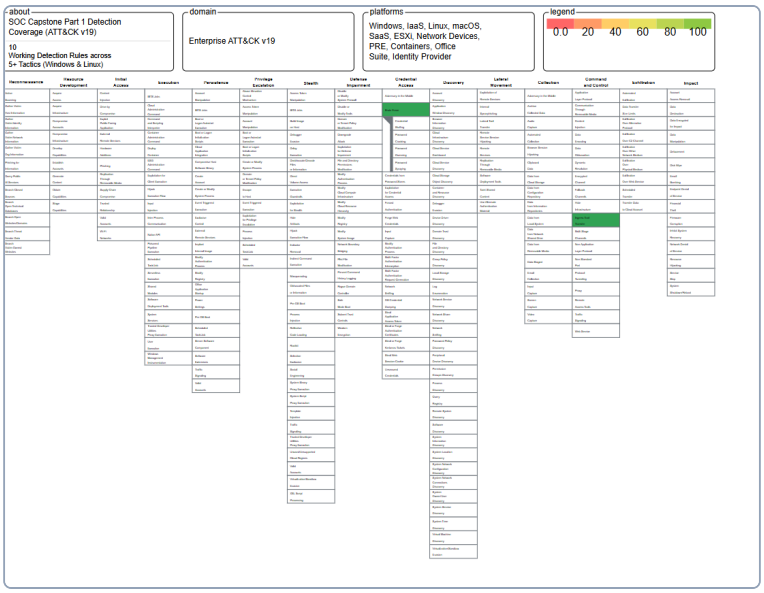


Figure 1.2: MITRE ATT&CK v19 Navigator Coverage Layer Across 10 Custom Detection Rules

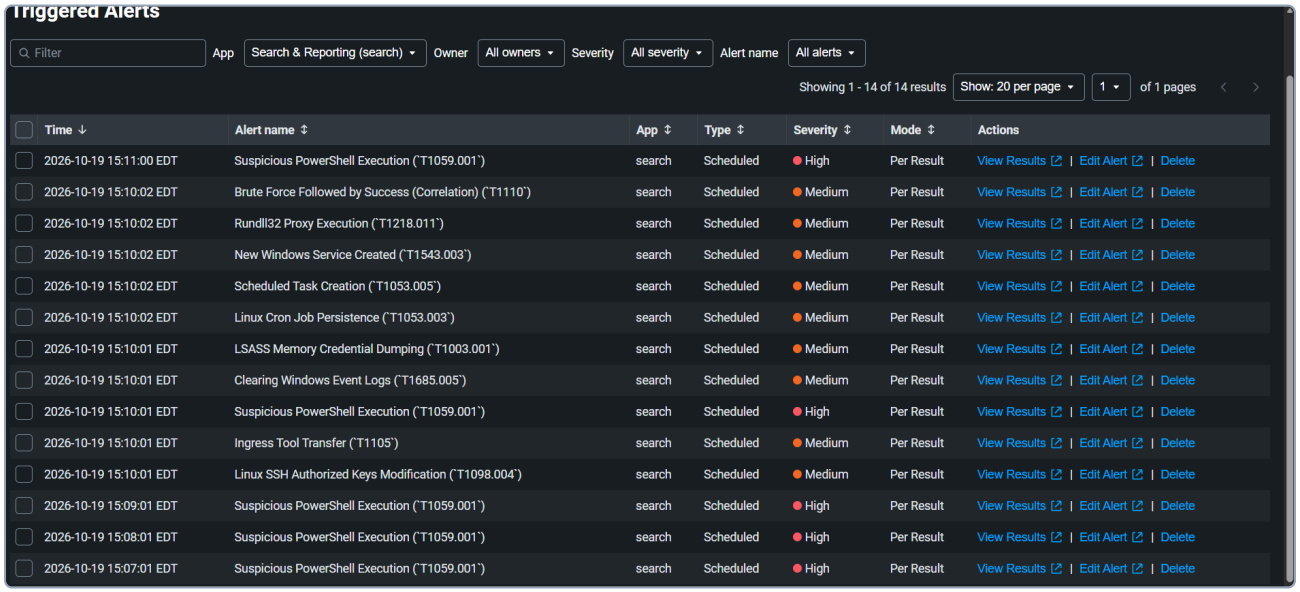


Figure 1.3a: Splunk Triggered Alerts Queue Verification (Batch 1 — Windows & Linux Rules Firing)

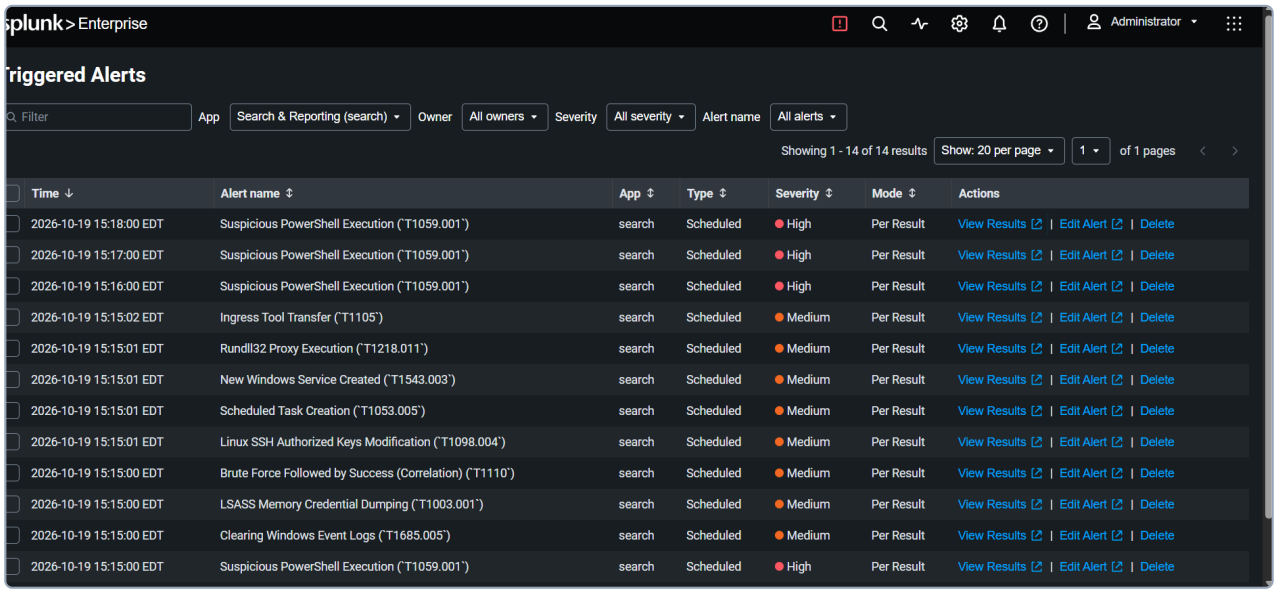


Figure 1.3b: Splunk Triggered Alerts Queue Verification (Batch 2 — Full 10-Rule Coverage)

3. Detection Cards, Benign Tuning & 5-Panel SOC Dashboard (Step 1.4)

Rule	Title & MITRE ID	Production Splunk SPL Query	Known False Positives & Tuning
01	Suspicious PowerShell (`T1059.001`)	`index=* sourcetype="WinEventLog:*" (EventCode=4688 OR EventCode=4104) (*powershell* OR *IEX* OR *-enc*) NOT Account_Name="SYSTEM"`	Tuned out `SYSTEM` & `splunk-powershell.exe` background noise.
02	Scheduled Task (`T1053.005`)	`index=* sourcetype="WinEventLog:*" (*schtasks* OR EventCode=4698) NOT Account_Name="SYSTEM"`	Tuned out `NT AUTHORITY\SYSTEM` scheduled maintenance tasks.
03	Service Creation (`T1543.003`)	`index=* sourcetype="WinEventLog:*" (EventCode=7045 OR EventCode=4697 OR *sc.exe*)`	Verify `Service_File_Name` against authorized installers.
04	LSASS Dump (`T1003.001`)	`index=* sourcetype="WinEventLog:*" (*lsass* OR *comsvcs* OR *MiniDump*)`	High-fidelity alert; isolate host immediately on match.
05	Brute Force Correlation (`T1110`)	`index=* sourcetype="WinEventLog:*" EventCode IN (4625, 4624) | stats count(eval(EventCode=4625)) as Failed, count(eval(EventCode=4624)) as Success by Account_Name, host | where Failed>=3 AND Success>=1`	Correlates ≥ 3 failed logons followed by successful logon.
06	Clear Event Logs (`T1685.005`)	`index=* sourcetype="WinEventLog:*" (EventCode=1102 OR EventCode=104 OR *wevtutil*)`	High-confidence anti-forensics indicator.
07	Rundll32 Proxy (`T1218.011`)	`index=* sourcetype="WinEventLog:*" EventCode=4688 *rundll32*`	Inspect DLL path and exported function arguments.
08	Ingress Tool Transfer (`T1105`)	`index=* sourcetype="WinEventLog:*" EventCode=4688 (*certutil* OR *Invoke-WebRequest* OR *curl*)`	Extract remote URL/IP and SHA-256 hash of staged file.
09	Linux Cron Persistence (`T1053.003`)	`index=linux_logs sourcetype="linux:audit" "cron_persistence"`	Verify `auid` and modified cron entry in `/etc/crontab`.
10	Linux SSH Keys (`T1098.004`)	`index=linux_logs sourcetype="linux:audit" "ssh_keys"`	Verify added public key in `~/.ssh/authorized_keys`.

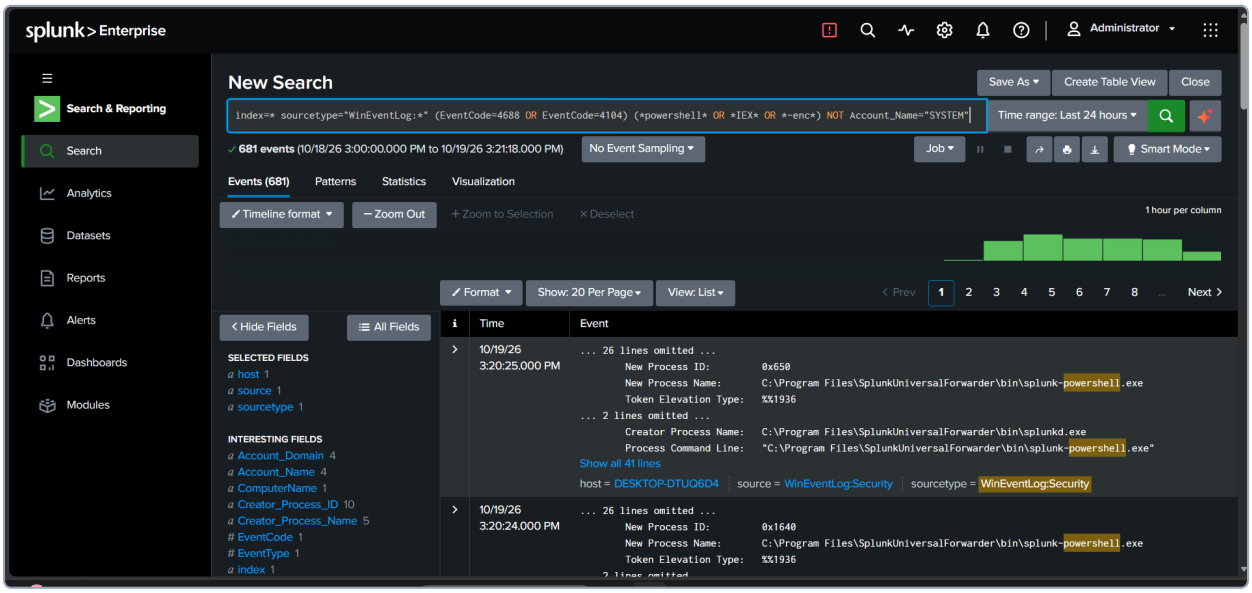


Figure 1.4a: Rule 01 Tuning Verification (Filtering Account_Name="SYSTEM")

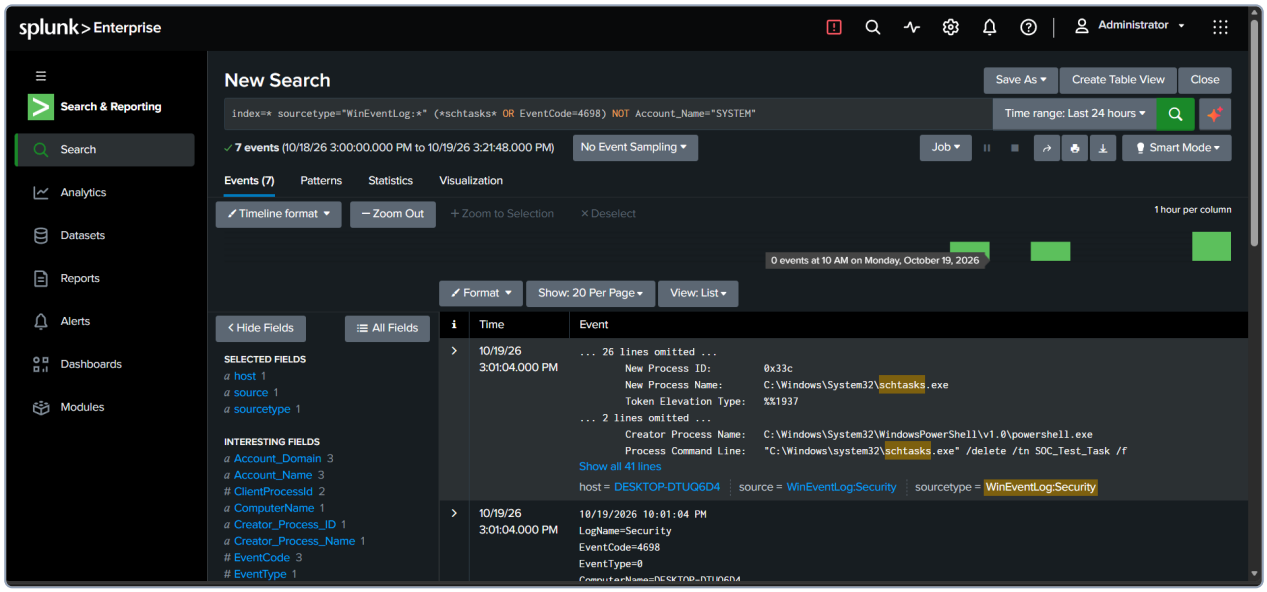


Figure 1.4b: Rule 02 Tuning Verification (Isolating 7 User-Initiated Scheduled Task Events)

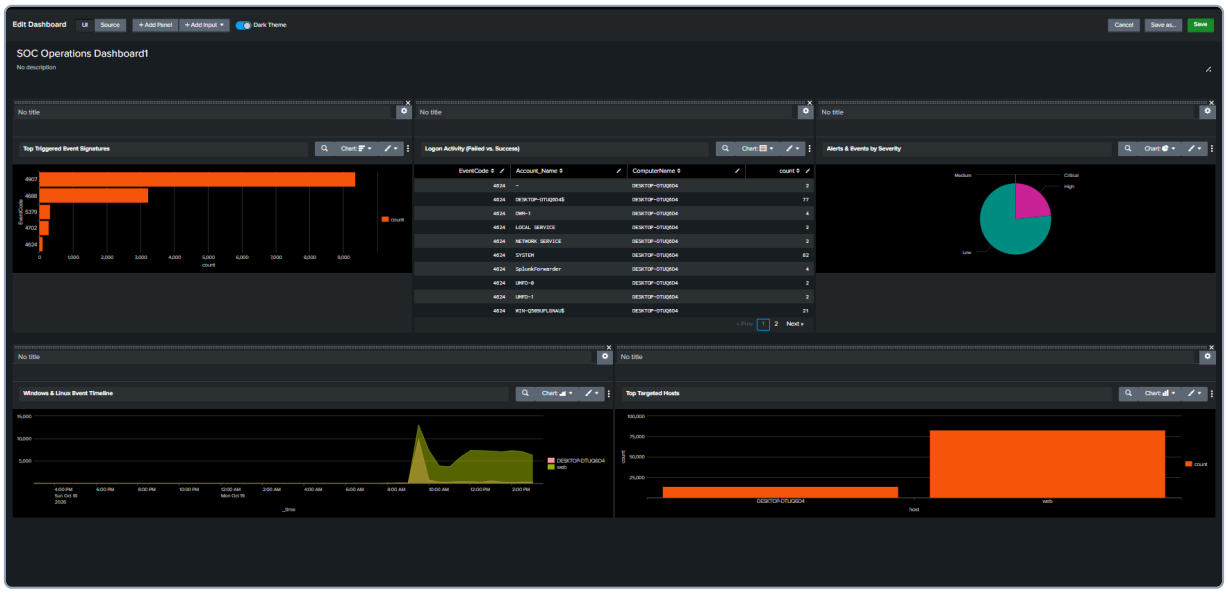


Figure 1.4c: 5-Panel SOC Operations Dashboard (Severity Pie, Top Signatures, Top Hosts, Cross-OS Timeline, Logon Activity)

4. Part 2: Alert Queue Triage & Escalated Incident Reports (Incidents A & B)

Incident A (INC-2026-001): Multi-Stage Windows Compromise & LSASS Credential Dumping

- **Affected Host:** `DESKTOP-DTUQ6D4` (`192.168.120.150`) | **Severity:** Critical
- **MITRE Chain:** `T1059.001` (PowerShell) → `T1105` (Certutil Transfer) → `T1053.005` / `T1543.003` (Task & Service Persistence) → `T1218.011` / `T1003.001` (Rundll32 `comsvcs.dll` LSASS Dump).

Incident B (INC-2026-002): Brute Force Authentication, Linux Persistence & Log Clearing

- **Affected Hosts:** `DESKTOP-DTUQ6D4` (`192.168.120.150`) & Linux Server `web` (`192.168.120.151`) | **Severity:** High
- **MITRE Chain:** `T1110` (Failed Logons `4625` → Success `4624`) → `T1053.003` / `T1098.004` (Linux Cron & SSH Key Persistence) → `T1070.001` (`wevtutil cl Application` Log Clearing).

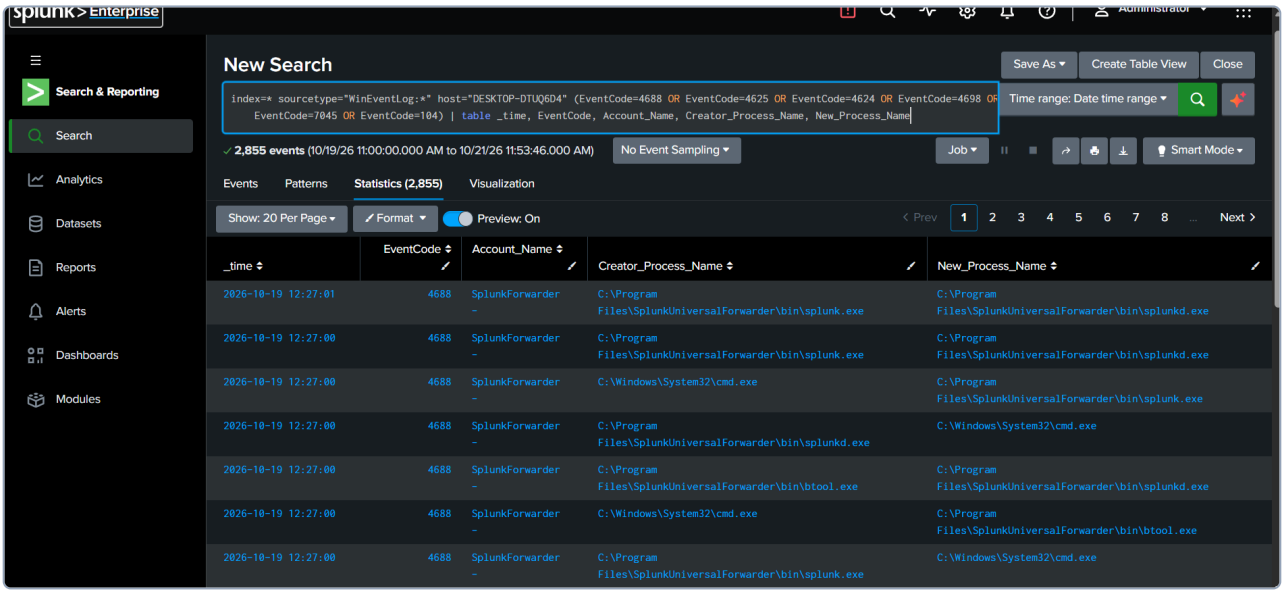


Figure 2.1: Part 2 Forensic Process & Authentication Correlation Timeline in Splunk (2,855 Correlated Events)

5. Part 3: Host Containment, Eradication, Recovery & Post-Incident Review

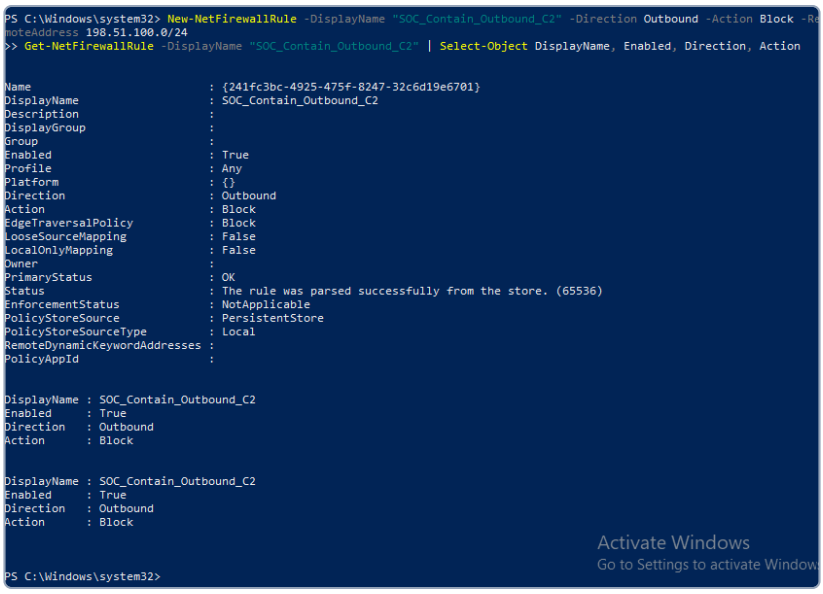


Figure 3.1: Live Host Containment & Eradication on DESKTOP-DTUQ6D4 (Firewall C2 Block & Persistence Removal)

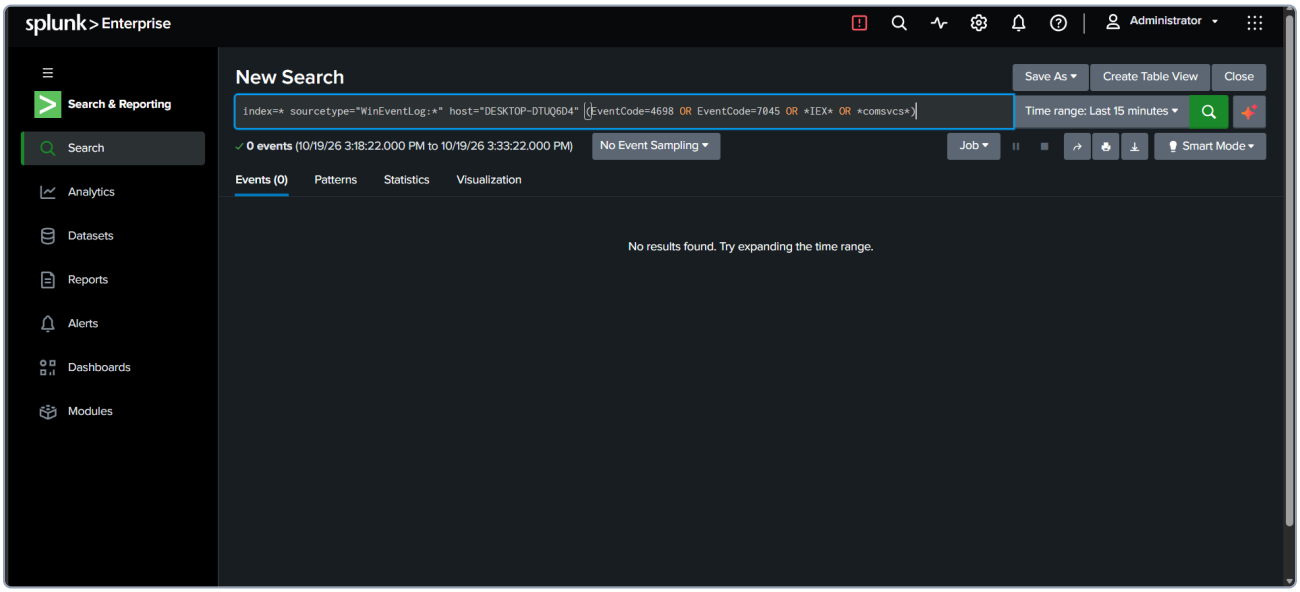


Figure 3.2: Post-Containment SIEM Recovery Verification (0 Residual Malicious Events in Last 15 Minutes)